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# Let's learn about **satellites**!

**Space mission** and **satellite** design course

# Orbits and Space Environment

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# Where does space begin?

Outer space is the vast region beyond Earth's atmosphere, extending between celestial bodies throughout the universe.

By international convention, space is considered to begin at the **Kármán Line**, located at an altitude of approximately 100 kilometers above sea level.

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# What is space made of?

At first glance, space may appear to be empty.
In reality, it is a **near-perfect vacuum**.

Space contains extremely low densities of particles, along with electromagnetic radiation, neutrinos, interplanetary dust, and cosmic rays.

The average background temperature of the Universe is about 2.7 Kelvin (**−270 °C**; −455 °F).

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# Space Environment

Space presents a harsh environment for both human life and satellite systems due to the presence of:

- Solar radiation
- Charged particle (plasma) environment
- Residual atmosphere (especially in low Earth orbit)
- Microgravity conditions
- Near-vacuum conditions
- Space debris

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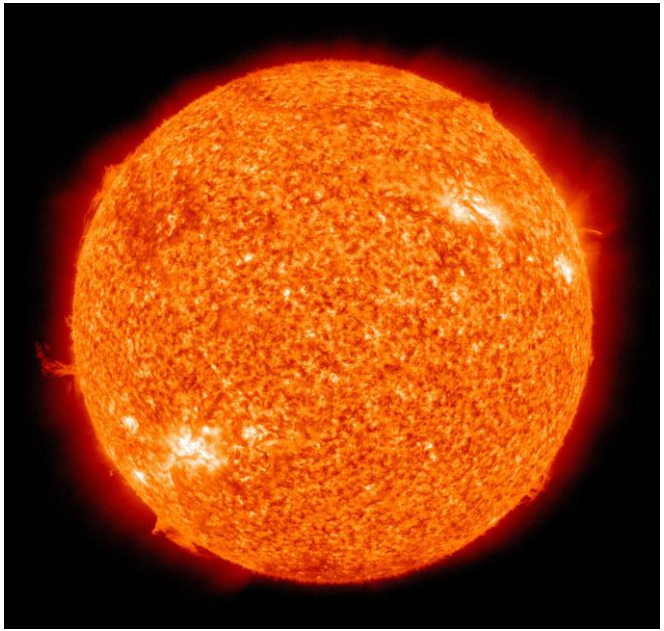
# Solar Radiation



The space environment in which satellites operate is strongly influenced by the Sun.

The Sun is a yellow dwarf star whose activity follows an approximately 11-year cycle.

It has a major impact on thermal design and power subsystems.

Solar radiation and charged particles emitted by the Sun can cause significant effects on spacecraft systems and pose risks to astronauts.

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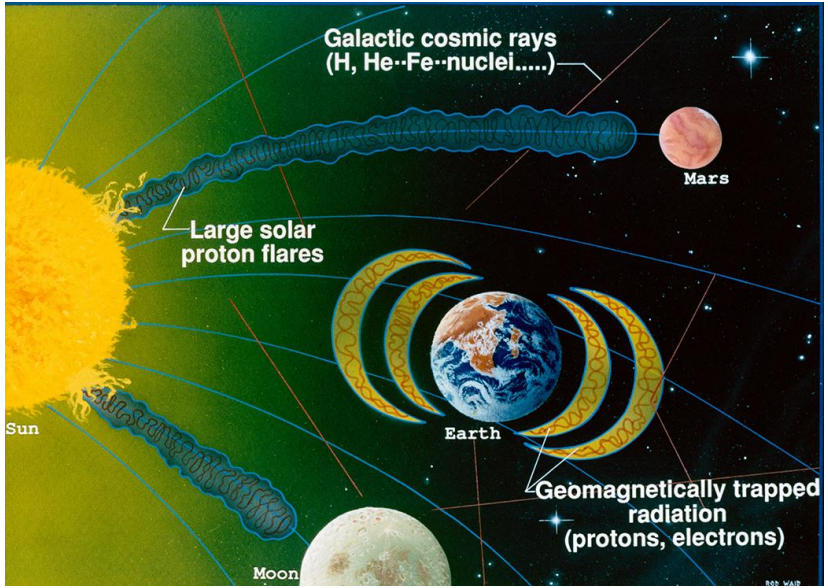
# Plasma Environment

- The Sun continuously emits a stream of charged particles (mainly electrons and protons), known as the solar wind, which extends far beyond the Solar
- The Earth's magnetic field shields us from most of these particles, but some become trapped, forming the **Van Allen** radiation belts.
- In addition, Galactic Cosmic Rays (GCRs) are high-energy particles originating outside the Solar System, further contributing to the harsh radiation environment in space.

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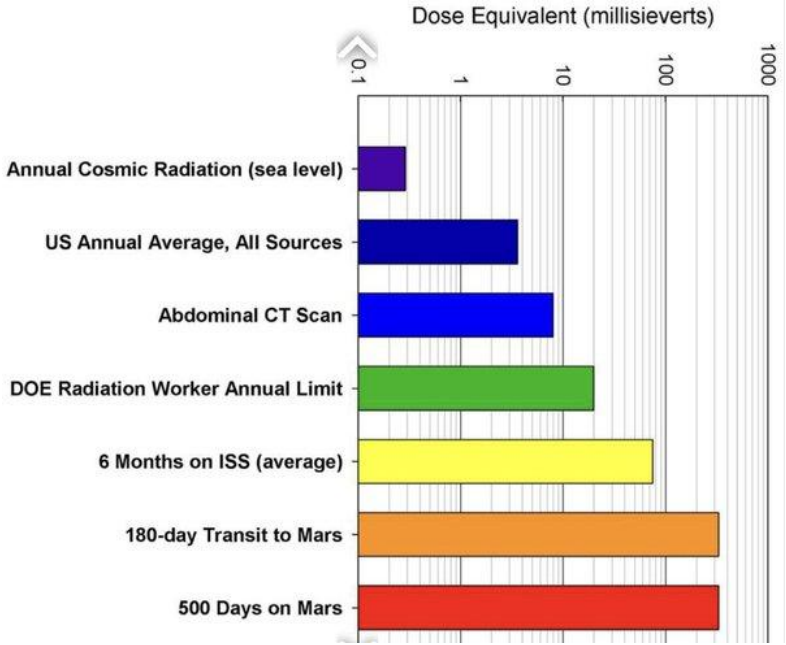
# Plasma Environment




- In addition to affecting onboard electronics, causing errors, degradation, and potential failure, high radiation levels can pose serious risks to human health.

- Prolonged exposure to radiation in space can lead to biological effects, and astronauts aboard the **ISS** are continuously monitored to assess radiation exposure and its potential health impacts.

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# Residual Atmosphere

The atmospheric density decreases with altitude, but it never becomes zero and remains significant at lower orbital altitudes.

The interaction between the residual atmosphere and a satellite generates atmospheric **drag**: a force that opposes the spacecraft's motion and gradually reduces its orbital altitude, eventually leading to re-entry.

Atmospheric density increases during periods of higher solar activity, resulting in stronger drag effects on satellites.

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# Microgravity

**Important**: Gravity does not disappear in space!

A common misconception is that orbit means zero gravity. In reality, gravity at typical low Earth orbit altitudes is still about 90% of its value at Earth's surface.

Microgravity occurs because the spacecraft and everything inside it are in continuous free fall around the Earth. Since all objects accelerate at the same rate, they appear weightless relative to each other.

In orbit, residual accelerations are typically on the order of $10^{-6}$ g, which is why the environment is called *microgravity*.

Under these conditions, physical, chemical, and biological processes behave differently compared to those on Earth.

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# Near Perfect Vacuum

In Earth orbit, ambient pressure is extremely low (approximately $10^{-5}$ Pa at ~400 km and around $10^{-15}$ Pa at GEO altitude ~36,000 km).

Under these near-vacuum conditions, volatile substances trapped inside materials can escape into space. This phenomenon is known as **outgassing**.

For this reason, only **space-qualified** materials can be used in spacecraft, and components must be tested in thermal-vacuum chambers before launch.

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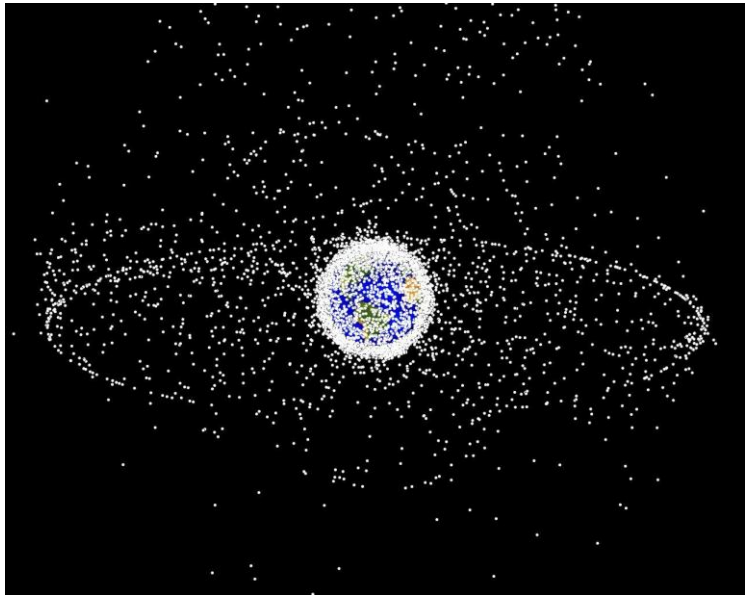
# Space Debris



Earth orbit is populated by a large number of natural and artificial objects, including small meteoroids, defunct satellites, spent rocket stages, and fragmentation debris.

These objects travel at very high velocities, making even small fragments potentially dangerous for operational spacecraft.

The growing debris population represents a critical challenge for the sustainability of current and future space activities.

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# Orbital Dynamics: Newton's Law

$$F = G\frac{m_1 m_2}{r^2}$$

The gravitational attraction is described by the **Newton's law of universal gravitation.** It states that every body attracts any other body in the Universe with a force that is proportional to the product of their masses and inversely proportional to the square of the distance between their centers.

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# The two-body problem

We model orbital motion by considering only **two bodies at a time** (e.g., Earth and a satellite). In reality, every object attracts every other object, but many influences are small enough to be neglected in a simplified model.

We also assume one mass is much smaller than the other (the satellite compared to Earth). Under these assumptions, the motion is governed by the fundamental two-body equation of orbital dynamics:

$$\frac{d^2\vec{r}}{dt^2} = -\frac{\mu}{r^2}\hat{r}$$

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# The two body problem

In this equation, **μ** depends only on the mass of the central body:

$$\mu = GM_{Planet}$$

This means that the gravitational acceleration experienced by a body in a gravitational field is independent of its own mass.

This principle was first demonstrated by **Galileo Galilei**, who showed that different masses fall with the same acceleration (neglecting air resistance).

Therefore, gravitational attraction acts as a central force, always directed toward the center of the attracting body.

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# The Keplerian Orbit

One of the solutions to the two-body problem is the elliptical orbit.

Planets in the Solar System - including Earth - orbit the Sun along elliptical paths.

The central body is located at one of the two foci of the ellipse.

According to **Kepler's second law**, the line joining the central body and the orbiting object sweeps out equal areas in equal times. As a result, the orbital velocity is higher at periapsis (closest point) and lower at apoapsis (farthest point).

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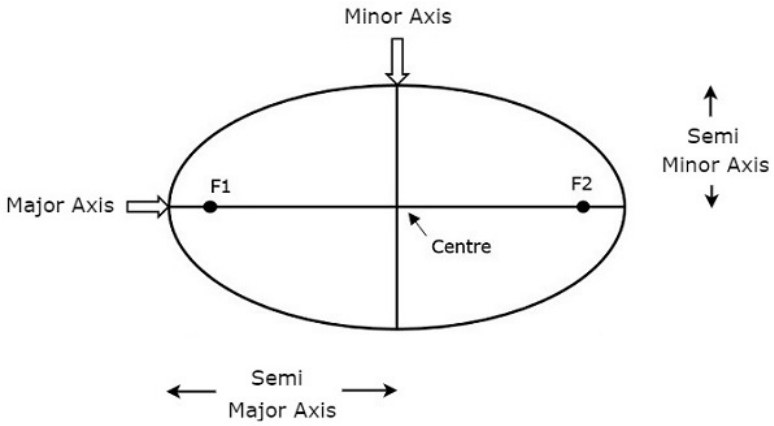
# Elliptical Orbit




There are two key parameters used to describe an elliptical orbit:

The semi-major axis, usually denoted as **a**, represents the size of the orbit and is used to compute quantities such as orbital period and energy.

The **eccentricity**, usually denoted as **e**, describes the shape of the orbit and ranges from 0 to 1.
If e is close to 0, the orbit is nearly circular; as e increases, the ellipse becomes more elongated.

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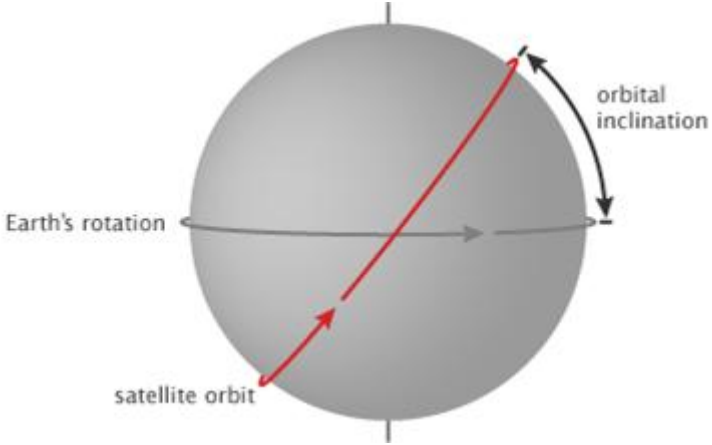
# Orbit Inclination

Another key orbital parameter is the **inclination**.

It is usually denoted by **i** and describes how much the orbital plane is tilted with respect to Earth's equatorial plane.

If the inclination is 90°, the orbit passes over the poles and is called a polar orbit.

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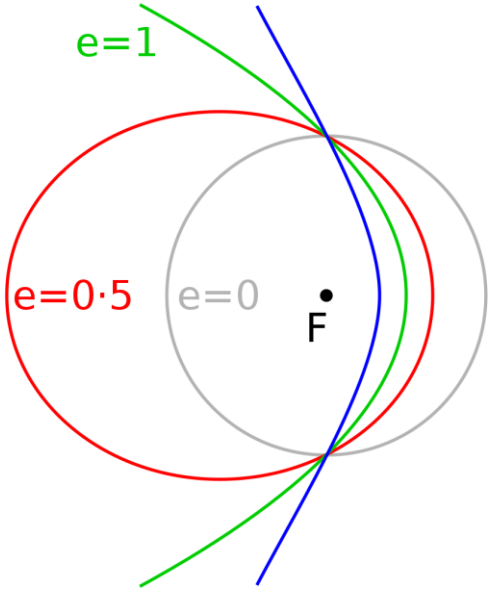
# More orbital shapes

If the eccentricity is equal to zero, the orbit is circular. Circular orbits are widely used in satellite applications.

If e = 1, the trajectory is parabolic, and if e > 1, it is hyperbolic.

Both cases correspond to escape trajectories and have important practical applications, which we will examine later.

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# Circular orbit

For a circular orbit, the orbital velocity is: $v_{circular} = \sqrt{\frac{\mu}{r}}$

This value is known as the **First Cosmic Velocity** (or circular orbital velocity).
It is the minimum velocity required to maintain a circular orbit around a central body.

Near Earth's surface (neglecting atmospheric drag), this velocity is approximately **7.9 km/s**.

The **orbital period**, or the time required to complete one full revolution, is:

$$T = 2\pi\sqrt{\frac{r^3}{\mu}}$$

It depends only on the orbital radius and the gravitational parameter of the central body.

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# Exercise - Circular orbit

$$\mu_{Earth} = 398604.3 \frac{km^3}{s^2}$$

$$R_{Earth} = 6371\ km$$

$$\mu_{moon} = 4904.9 \frac{km^3}{s^2}$$

$$R_{Moon} = 1737.4\ km$$

$$\mu_{mars} = 42828.4 \frac{km^3}{s^2}$$

$$R_{Mars} = 3389.5\ km$$

Given the values on the left, compute:

1. The circular velocity and the period of a satellite in a circular orbit around Earth at 550 km of altitude.
2. The circular velocity and the period of a satellite in a circular orbit around Earth at 1600 km of altitude.
3. The circular orbit and the period of a satellite in a circular orbit around the Moon at 550 km of altitude.
4. The circular orbit and the period of a satellite in a circular orbit around mars at 550 km of altitude.

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# Solutions - Circular Orbits

1. Velocity = 7.59 km/s; T = 5730 s = 95.5 min = 1.59 hours

1. Velocity = 7.07 km/s; T = 7082 s = 118 min = 1.97 hours

1. Velocity = 1.464 km/s; T = 9814 s = 163 min = 2.72 hours

1. Velocity = 3.30 km/s; T= 7507 s = 125 min = 2.09 hours

From this exercise, we can see that the more massive the planet, the higher the velocity required for a spacecraft to maintain a stable orbit. Furthermore, the closer a spacecraft is to the planet, the greater its orbital velocity.

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# Parabolic Orbit

$$v_{escape} = \sqrt{2\frac{\mu}{r}} = \sqrt{2}v_{circular}$$

If e = 1, the trajectory is parabolic and therefore not closed. In this case, a spacecraft reaches the minimum energy needed to escape the gravitational influence of the central body (neglecting other forces).

This concept is relevant for interplanetary travel: for example, to go from Earth to Mars, a spacecraft must first escape Earth's gravity well.

This is the **Second Cosmic Velocity,** also known as **Escape Velocity**.

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# Hyperbolic Orbit

While a parabolic trajectory allows a spacecraft to escape a planet, it has zero velocity at infinity.

For interplanetary missions, we require a hyperbolic trajectory, where the eccentricity e>1 and the velocity at infinity is greater than zero.

The velocity at infinity, denoted $v_\infty$, is called the **Hyperbolic Excess Velocity**.

It is related to the spacecraft velocity by: $v_\infty^2 = v^2 - v_{escape}^2$

This quantity represents the residual speed of the spacecraft once it has escaped the gravitational influence of the central body.

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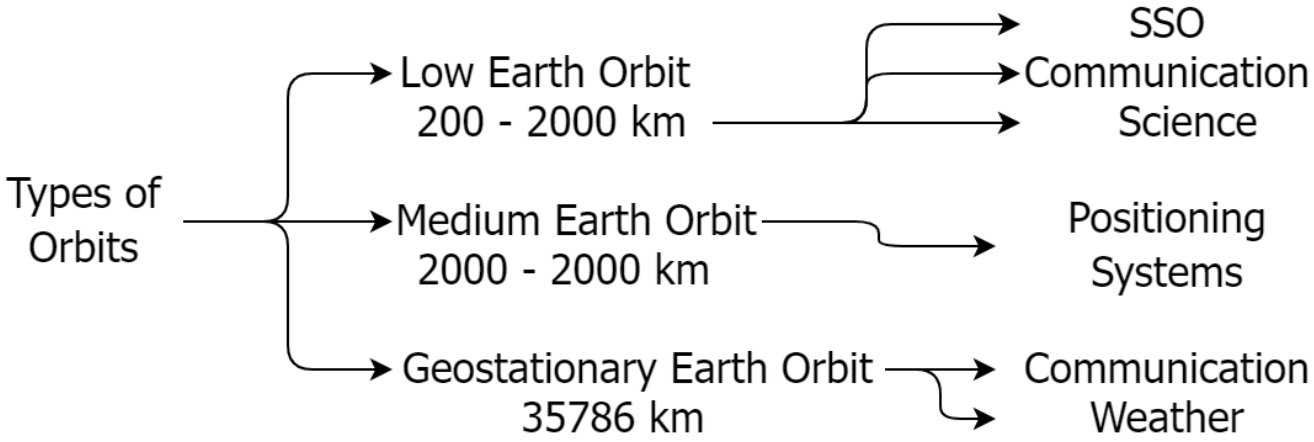
# Mission Types and Orbits

The mission types around Earth could be divided into different categories based on the **orbit altitude**. In this way, we can talk about Low Earth Orbit (LEO), Medium Earth Orbits (MEO), and Geostationary Earth Orbit (GEO). Every orbit has its own uses and practical applications.

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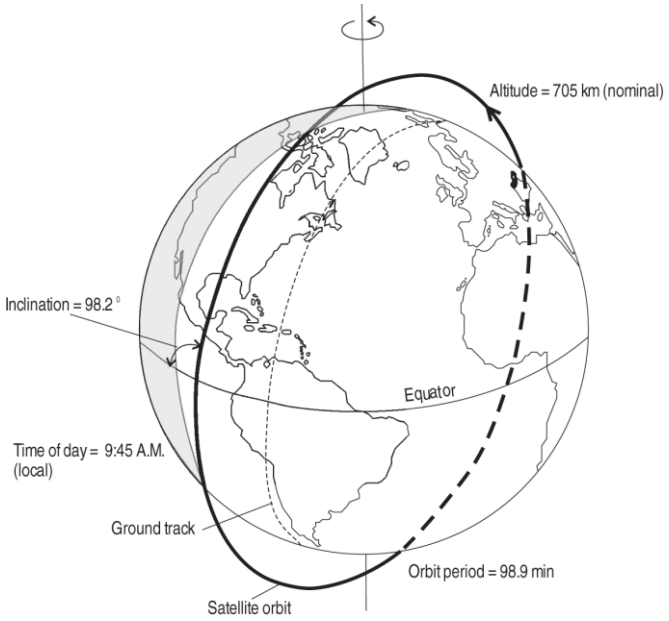
# Low Earth Orbits




Low Earth Orbits are the most commonly used orbital regime.

They require less energy to reach compared to higher orbits, making launch costs lower.
LEOs are particularly suitable for a wide range of applications, including Earth observation, telecommunications, and scientific missions.

A specific class of near-polar LEOs, known as **Sun-Synchronous Orbits (SSO)**, maintains a nearly constant angle with respect to the Sun. This ensures consistent lighting conditions, which is especially important for remote sensing and imaging missions.

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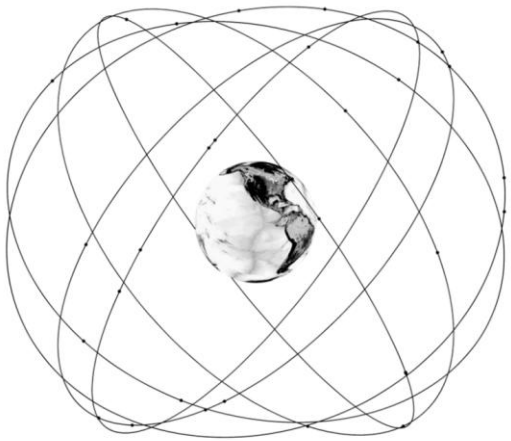
# Medium Earth Orbit

MEOs are predictable and reliable orbits often used for **global navigation satellite systems** (GNSS). GPS, GLONASS, Galileo, and BeiDou operate in MEO.

Some communications missions also use highly elliptical orbits (HEO), such as Molniya and Tundra.

These trajectories keep the spacecraft moving more slowly near apogee, providing long dwell time over high latitudes and making them useful for coverage in Arctic regions.



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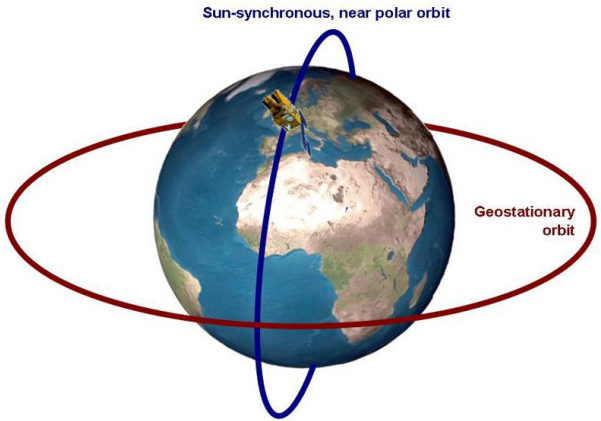
# Geostationary Orbit




At an altitude of about 35,786 km above the equator, the orbital period matches Earth's rotational period, corresponding to one day (approximately 23 hours, 56 minutes).

As a result, a **geostationary** (GEO) satellite remains fixed over the same point on Earth's surface. This makes GEO satellites particularly well suited for communications, broadcasting, and weather monitoring applications.

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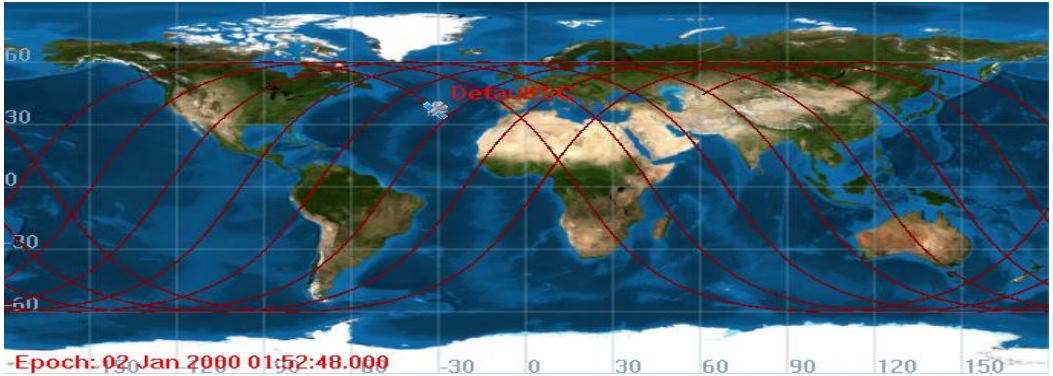
# Ground Track of a satellite

It's extremely relevant to know where a satellite is passing over Earth: you need to know when it passes over the ground stations, or which parts of the Earth you will be able to remote sense.

This is usually visualized by the **Ground Track**.
https://satellitetracker3d.com/

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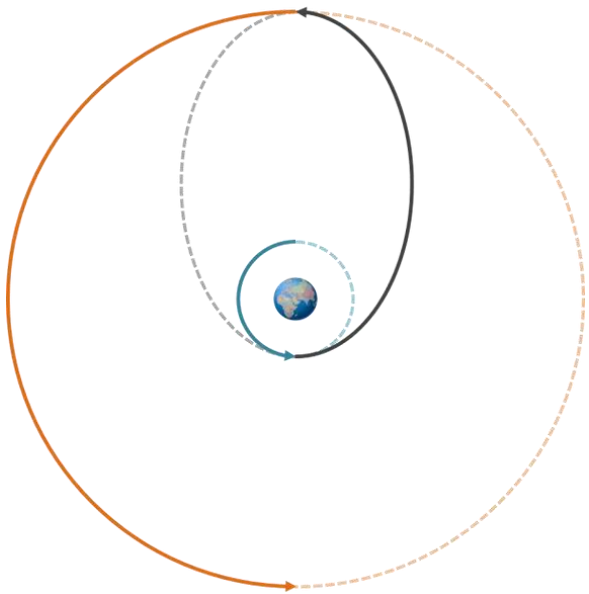
# Orbital Transfers



We have seen LEO, MEO, and GEO but how do we move from one orbit to another?

To perform an **orbital transfer**, we use one or more transfer orbits: trajectories that share one point with the initial orbit and one point with the target orbit.

In the figure, you can see a **Hohmann transfer**. In this case, the velocity changes (burns) are performed at perigee and apogee of the transfer orbit.

The Hohmann transfer is the most fuel-efficient transfer between two coplanar, circular orbits when impulsive burns are assumed.

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# Orbital Perturbations

In reality, orbital motion is not as simple as the ideal two-body model.

All orbiting bodies are affected by additional forces known as orbital perturbations. Therefore, a real orbit is not a perfect Keplerian ellipse, but a trajectory that continuously evolves due to several perturbing effects.

The main perturbations include:

- **Earth's non-uniform gravitational potential** (e.g., J2 effect and higher harmonics)
- **Solar radiation pressure**
- **Atmospheric drag**
- Third-body perturbations (Sun, Moon)

In practice, orbital mechanics is more complex than the simplified models we have seen so far.

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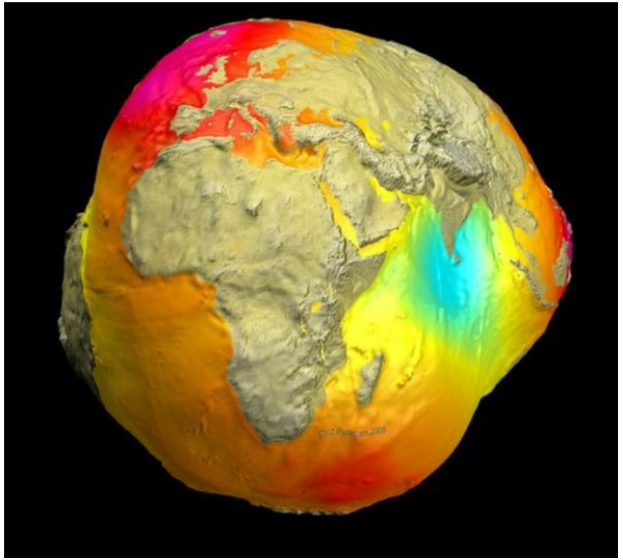
# Gravitational Potential of the Earth



Earth is not a perfect sphere, so the simple Keplerian model is only an approximation.

Its gravitational field is described using spherical harmonics, a series of coefficients that account for mass irregularities.

Some coefficients, such as J2 and J22, are so important that the perturbations they cause are commonly referred to by their names.

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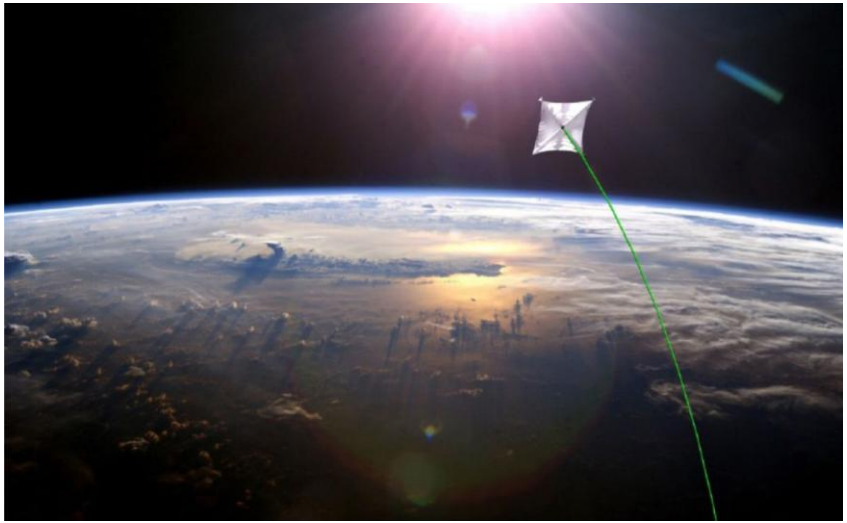
# Solar Radiation Pressure

Photons striking a satellite carry momentum, and part of it is transferred to the spacecraft, slightly changing its momentum.

This effect, known as **Solar Radiation Pressure**, can modify orbital parameters and must be taken into account.

Although the force is small, it can be exploited for propulsion using solar sails.



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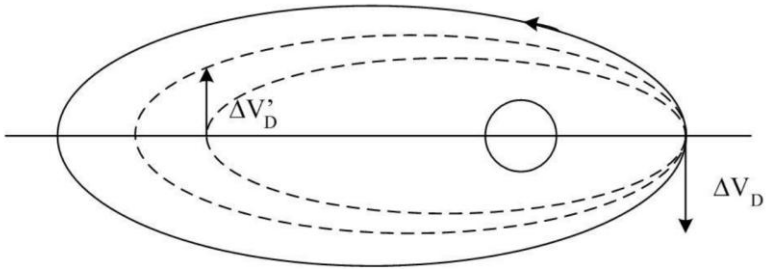
# Atmospheric Drag

**Atmospheric drag** has two main effects:

First, it tends to circularize the orbit: a highly elliptical orbit will gradually become more circular if its perigee is low enough to interact with the upper atmosphere. This effect is sometimes exploited after atmospheric capture around planets such as Mars.

Second, drag reduces the semi-major axis over time, causing orbital decay and eventually re-entry. This can be beneficial at end of life, but during operations it may require station-keeping maneuvers to maintain altitude.

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